

Mathematical Statistics (I)

Solutions to the Final Exam.

June 18, 2018

In the followings, $\bar{X} = \sum X_i/n$ and $s^2 = \sum (X_i - \bar{X})^2/(n-1)$ denote the sample mean and the sample variance, respectively.

1.(100) Let (X, Y) follow the bivariate normal distribution with $E(X) = E(Y) = 0$, $Var(X) = Var(Y) = 1$, $Cov(X, Y) = \rho$. (Hint : The pdf of $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is $(2\pi)^{-p/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\}$).

(1)(50) Derive the marginal pdf of X .

(Sol) By the given hint, it is easy to show that the jpdf is $\frac{1}{(2\pi)\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)\}$. Hence, the marginal pdf of X is $\int \frac{1}{(2\pi)\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)\} dy = e^{-y^2/2} \int \frac{1}{(2\pi)\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x - \rho y)^2\} dy = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}$

(2)(50) Compute the conditional mean and variance of Y given X .

(Sol) Can compute that $\frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(y - \rho x)^2\}$. Therefore, the conditional mean and variance are ρx and $1 - \rho^2$, respectively.

2.(50) Show that the sample correlation coefficient is a consistent estimator of the population correlation coefficient.

(Sol) The sample corr. coeff. is $r_n = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$ and $\frac{1}{n} \sum (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{1}{n} \sum X_i Y_i - \bar{X} \bar{Y} \rightarrow E(XY) - E(X)E(Y) = Cov(X, Y)$ by WLLN and the continuous mapping principle. Also, $\frac{1}{n} \sum (X_i - \bar{X})^2 = \frac{1}{n} \sum X_i^2 - \bar{X}^2 \rightarrow E(X^2) - E^2(X) = Var(X)$ and $\frac{1}{n} \sum (Y_i - \bar{Y})^2 = \frac{1}{n} \sum Y_i^2 - \bar{Y}^2 \rightarrow E(Y^2) - E^2(Y) = Var(Y)$. Finally, $r_n \rightarrow \rho$ by the continuous mapping principle.

3.(50) Let $X_1, X_2, \dots, X_n$ be r.s. from $N(0, 1)$. Find a limiting distribution of $\sqrt{n}(\frac{1}{n} \sum X_i^2 - 1)$.

(Sol) Note that $X_i^2 \sim \chi^2(1)$, so that, by CLT, $\sqrt{n}(\frac{1}{n} \sum X_i^2 - 1)/\sqrt{2} \rightarrow N(0, 1)$. Hence, the limiting distribution of $\sqrt{n}(\frac{1}{n} \sum X_i^2 - 1)$ is $N(0, 2)$.

4.(50) Let $X_1, X_2, \dots, X_n$ be r.s. from $\Gamma(\alpha, \beta)$. Show that $\bar{X}^2/s^2$ is a consistent estimator of α .

(Sol) By WLLN, $\bar{X} \rightarrow E(X_1) = \alpha\beta$, so that $\bar{X}^2 \rightarrow \alpha^2\beta^2$. Also, $s^2 \rightarrow \text{Var}(X_1) = \alpha\beta^2$. Therefore, $\bar{X}^2/s^2 \rightarrow \alpha$.

5.(50) Let X be a r.v. with cdf $F(\cdot)$. Find the distribution of $F(X)$ using the cdf technique.

(Sol) Let $Y = F(X)$, then the cdf of Y is $G(y) = P(F(X) \leq y) = P(X \leq F^{-1}(y)) = F(F^{-1}(y)) = y$. Therefore, the distribution of $F(X)$ is $U(0, 1)$.

6.(100) Let $X_1, X_2, \dots, X_n$ be r.s. from $U(0, 1)$.

(1)(50) Find the distribution of $-\log X_1$ using the cdf technique.

(Sol) Let $Y_i = -\log X_1$, then the cdf of Y is $G(y) = P(-\log X \leq y) = P(X > e^{-y}) = 1 - e^{-y}$, which is the cdf of $\Gamma(1, 1) = \text{Exp}(1)$.

(2)(50) Show that the limiting distribution of $\sqrt{n}(-\frac{1}{n} \sum \log X_i - a)$ is $N(0, b^2)$, and find a and b .

(Sol) By CLT, $\sqrt{n}(-\frac{1}{n} \sum \log X_i - E(Y_1))/\sqrt{\text{Var}(Y_1)} = \sqrt{n}(-\frac{1}{n} \sum \log X_i - 1)/\sqrt{1} \rightarrow N(0, 1)$. Hence, $a = b = 1$.

7.(50) Let $X_1, X_2, \dots, X_n$ be r.s. from $N(\mu, \sigma^2)$. Find k s.t. $E(ks) = \sigma$, where $s = \sqrt{s^2}$ is a sample standard deviation.

(Sol) Note that $v \equiv (n-1)s^2/\sigma^2 \sim \chi^2(n-1)$. Hence, $E(s) = E(\sqrt{v}\sigma/\sqrt{n-1}) = \frac{\sigma}{\sqrt{n-1}} \int v^{1/2} \frac{v^{\frac{n-1}{2}-1} e^{-v/2}}{\Gamma(\frac{n-1}{2}) 2^{\frac{n-1}{2}}} dv = \frac{\sigma}{\sqrt{n-1}} \frac{\Gamma(\frac{n}{2})\sqrt{2}}{\Gamma(\frac{n-1}{2})}$. Hence, $k = \frac{\Gamma(\frac{n-1}{2})\sqrt{n-1}}{\Gamma(\frac{n}{2})\sqrt{2}}$.

8.(50) Let $F \sim F(1, r)$ and $T \sim t(r)$. Also, let $P(F > F_\alpha(1, r)) = P(T > t_\alpha(r)) = \alpha$. Show that $F_\alpha(1, r) = t_{\alpha/2}^2(r)$.

(Sol) Note that $\alpha = P(F > F_\alpha(1, r)) = P(|T| > t_{\alpha/2}(r)) = P(T^2 > t_{\alpha/2}^2(r))$. Therefore, we must have $F_\alpha(1, r) = t_{\alpha/2}^2(r)$ because $T^2(r) = F(1, r)$.

Summary Statistics on Final Exam Scores

$n = 69$, average = 153.2

Top Seat : 임경태, 정상훈

Summary Statistics on Total Scores

$n = 69$, average = 403.9

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